

# Employee Management

## Overview

An Employee Management application built with ASP.NET Core MVC, .NET Web API, and SQL Server. Includes an NUnit test project for integration and unit testing.

It allows users to perform CRUD operations on employee data through a web interface and API endpoints.

## Project Structure

- ❖ EmployeeManagement -Solution
  - EmployeeManagement – **Presentation Layer**
    - ◆ Employee Controllers – Handle UI Requests
    - ◆ Views
      - Index – Employee List Page
      - CreateEmployee – Create /Edit (Employee creation and listing are handled asynchronously (no page reload))
    - ◆ Appsettings.json – Db Configuration for web
  - EmployeeManagementAPI - **API Layer**
    - Employee Controllers – API Endpoints
  - EmployeeData - **Data Access Layer**
    - EmployeeDbContext
      - ◆ EmployeeDbContext – Database Context
    - EmployeeModel – Table models
    - ResponseModel – Request response model
  - EmployeeServices - **Business Logic Layer**
    - Employee
      - ◆ EmployeeService – Employee CRUD operation service/Interface
      - ◆ HelperFunctions – Validation helper
  - EmployeeTest - **Testing Layer**
    - API -API integration Test Setup
    - Web - WEB integration Test Setup

## Technologies Used

- ASP.NET Core MVC
- .NET Web API
- SQL Server
- Entity Framework Core
- AJAX
- Bootstrap
- NUnit (Testing)
- Moq (Mocking)
- ASP.NET Core MVC Testing

## Run the Application Locally

Prerequisites:

- Visual Studio 2022/2026
- SQL Server

Startup Configuration

Set Multiple Startup Projects:

- EmployeeManagement
- EmployeeManagementAPI

Run the Application

- Start solution

URLs:

- Web Application (Frontend)  
https://localhost:7211
  - Opens Employee CRUD UI
- API Base URL  
https://localhost:7224
  - Use this in Postman to test APIs

## API Testing (Postman)

A sample Postman collection is included in: /PostmanCollection

Sample API Request

**URL:** <https://localhost:7224/api/Employee/CreateEmployee>

**Method:** Post

**Body(Json):**

```
{  
  "employeeId": 0,  
  "employeeName": "finny johnn",  
  "designation": "Software Developer",  
  "emailAddress": "john@test.com",  
  "mobilePhone": "65545454",  
  "dateOfBirth": "1996-06-15",  
  "nationality": "indian",  
  "isActive": true  
}
```

**Running Tests**

Open Test Explorer in Visual Studio:

Test → Test Explorer

- Run all tests before starting the application
- The test project includes:
  - API integration tests
  - Web integration tests

Results will be displayed directly in Test Explorer